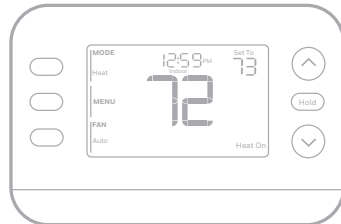




X2P Programmable Thermostat RTH20B/RTHC20B (1H/1C) RTH21B/RTHC21B (1H/1C) RTH22B/RTHC22B (2H/2C conventional or 2H/1C HP)

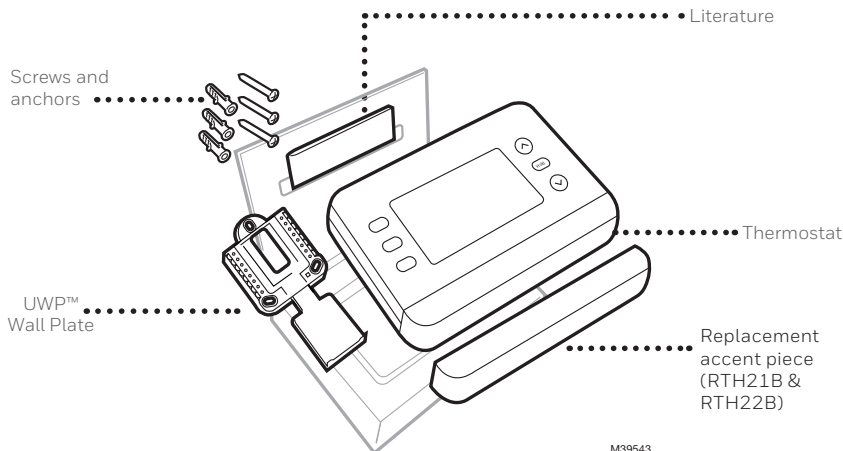


Installation and User Guide

This document contains the following sections:

- 1) Installation and Wiring
- 2) Configuration
- 3) System Operation
- 4) Accent Piece Replacment
- 5) Troubleshooting

Package includes:



Tools you will need



Phillips screwdriver



Small flat head screwdriver



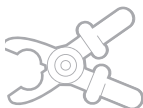
Pencil



Level

M36270

Tools you may need



Wire stripper



Needle-nose pliers



Drill and
drill bit

READ AND SAVE THESE INSTRUCTIONS

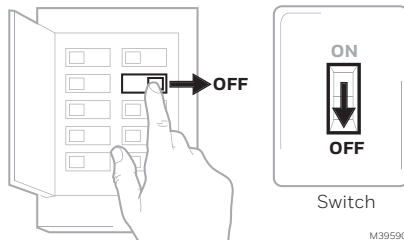
Installation and Wiring

Removing the old thermostat

1 Turn power OFF.



To protect yourself and your equipment, turn off the power at the breaker box or switch that controls your heating/cooling system. Some systems may have separate heating and cooling breakers.



Breaker box

M39590

2 Check that your system is off.



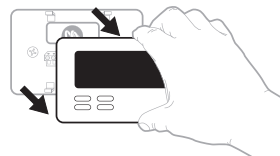
Change the temperature on your old thermostat to be above room temperature in heat mode or below it in cool mode. If you don't hear the system turn on within 5 minutes, the power is off. **Note:** If you have a digital thermostat that has a blank display, skip this step.



3 Remove the old thermostat's faceplate.

On most thermostats, you can take off the faceplate by grasping and gently pulling. Some thermostats may have screws, buttons, or clasps.

Do not remove any wires from your thermostat at this time!



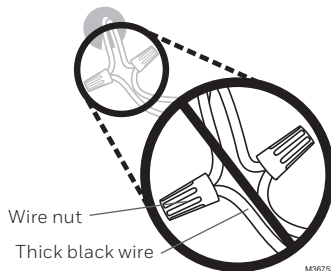
- 4 Make sure there are no 120/240V wires.**

Do you have thick black wires with wire nuts?



Is your thermostat 120V or higher?

If you answered yes to either of these questions, you have a line voltage system and the thermostat will not work.



- 5 Take a picture of how your wiring looks right now.**

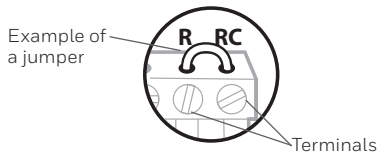
Be sure to include the letters next to the terminals where the wires are inserted. This will be a helpful reference when wiring your thermostat.

Tip: If the color of your wires has faded or if 2 terminals have the same wire color, use the wire labels provided in the package to label each wire.



- 6 Make note of any jumpers**

A jumper connects one terminal to another terminal. It may look like a small staple or even a colored wire.



- 7 Record whether you have wires in the following terminals.**
Do not include jumpers as a part of your count. This thermostat does not need jumpers.

Terminal	Wire Color
☐ R	_____
☐ RH	_____
☐ Rc	_____

- 8 Write down the color of the wires.**
Check mark the wires that are connected to terminals. Next to the check mark, write down the color of the wire. **Do not include jumpers as a part of your count.**

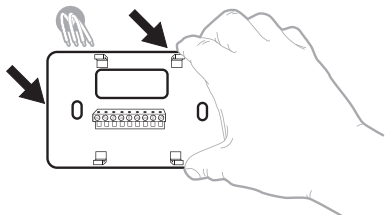
Check all that apply (Not all will apply):

Terminal	Wire Color	Terminal	Wire Color
☐ Y	_____	☐ A or L/A	_____
☐ Y2	_____	☐ O/B	_____
☐ G	_____	☐ W2 or AUX	_____
☐ C	_____ required _____	☐ E	_____
		☐ W	_____
		☐ K	_____

M39554

- 9 Disconnect the wires and remove the old wall plate.**
Use a screwdriver to release wires from terminals. Then, use a wire label to identify each wire as it's disconnected. The letter on the wire label should match the letter on the terminal.

Tip: To prevent wires from falling back into the wall, wrap the wires around a pencil.



Wire Labels		Étiquettes de fils		Etiquetas para los cables					
Add wire label to each wire as you disconnect it from the terminal.		Ajoutez une étiquette à chaque fil dès que vous le déconnectez du terminal.		Coloque une etiqueta en cada cable al desconectarlo del terminal.					
Do not disconnect any wires unless it corresponds to the terminal.		Ne déconnectez aucun fil à moins qu'il corresponde au terminal.		No desconecte ningún cable a menos que corresponda al terminal.					
B	B	Y2	Y2	C	C	E	E	F	F
G	G	H	H	L	L	O	O	P	P
R	R	RC	RC	RH	RH	T	T	U	U
V/R	V/R	W	W	W1	W2	W2	W3	W3	W3
X	X	X1	X1	X2	X2	Y	Y	Y1	Y1

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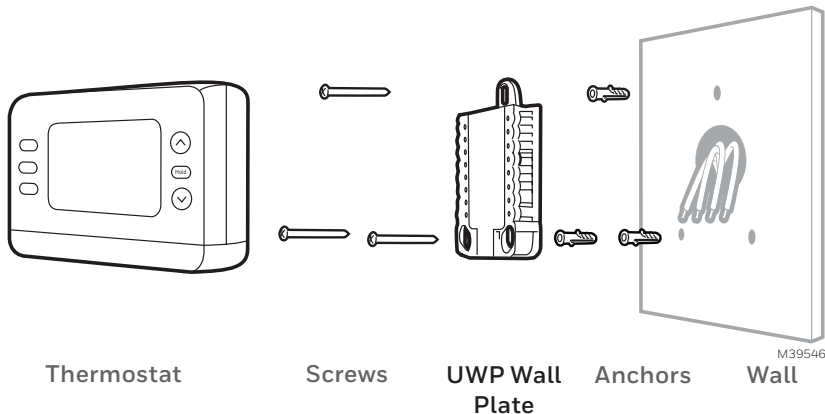
www.resideo.com

644030375 • 5403 • Printed in U.S.A.

www.millbro.com

M281008

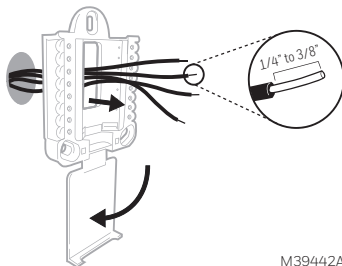
Installing the thermostat



1 Bundle and insert wires through the UWP.

Pull open the UWP and insert the bundle of wires through the back of the UWP.

Make sure at least **1/4-inch** of each wire is exposed for easy insertion into the wire terminals.

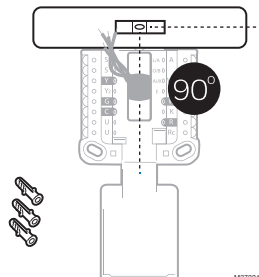


2 Insert the wall anchors.

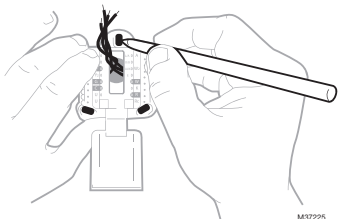
It is recommended that you use the wall anchors included in the box to mount your thermostat.

You can use the **UWP** to mark where you want to place the wall anchors.

- Level the wall plate.
- Mark the location of the wall anchors using a pencil.
- Using a 3/16" bit, drill the holes.
- Insert wall anchors.
- Make sure anchors are flush with wall.



- 3 Set R-switch position and insert R-wire or wires.**
Set the R-switch up or down based on your wiring notes in **Step 7** of the section **Removing the Old Thermostat**.

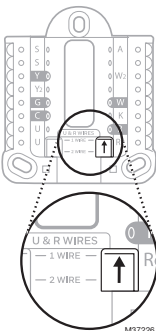


M37225

Insert wires into the inner holes of the terminals on the UWP. The tabs will stay down once the wire is inserted.

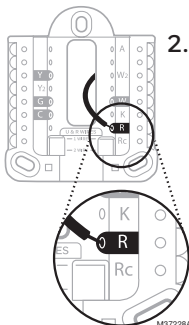
NOTE: Alternate wiring options are shown on the following pages

If you have 1 R-wire (R,Rh, or Rc)



M37226

- 1. Set R-switch to the up position.**

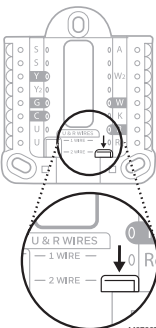


M37226A

- 2. Insert your R-wire (R, Rh or Rc) into R-terminal.**

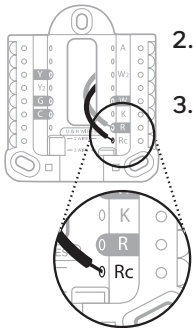
or

If you have 2 R-wires (R or Rh, and Rc)



M37227

- 1. Set R-switch to the down position.**



M37229A

- 2. Insert your Rc wire into Rc-terminal**
3. Insert your R or Rh wire into R-Terminal.

Wiring

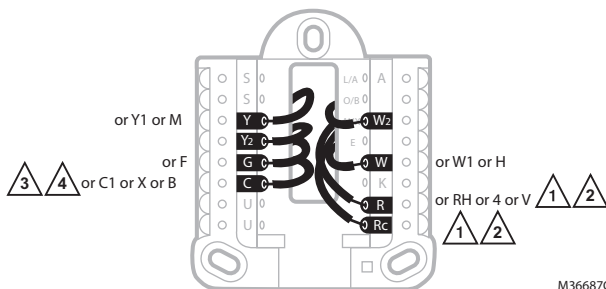
Conventional Systems

Alternate wiring (conventional systems)

If labels do not match terminals, connect wires as shown below (see notes below).

WIRING NOTES:

1. If you must connect both R and Rc wires, set the R Slider Tab to the down position (2 wires).
2. If your old thermostat had both R and RH wires, set the R Slider Tab to the down position (2 wires). Then connect the R wire to the Rc terminal, and the RH wire to the R terminal.
3. If your old thermostat had only 1 C or C1 wire, connect it to the C terminal. If your old thermostat had 2 C or C1 wires, wrap each separately with electrical tape and do not connect them.
4. C does not power the thermostat display or operations; batteries are always required.



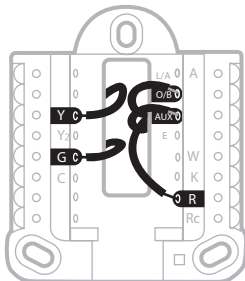
M36687C

Heat Pump

WIRING NOTES:

1. Match each labeled wire with same letter on new thermostat.
2. Insert the wires into the matching terminal.

NOTE: If you have difficulty inserting wires, you may have to press down the terminal push button next to the corresponding terminal. If the labels do not match the letters on the thermostat, see the diagram above.



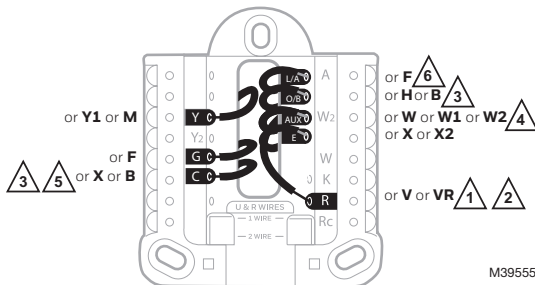
Heat Pumps

Alternate wiring (for heat pumps only)

WIRING NOTES:

1. Keep R Slider Tab in the **up** position (1 wire).
2. If your old thermostat had both **V** and **VR** wires, stop now and contact a qualified contractor for help.
3. If your old thermostat had separate **O** and **B** wires, attach the **B** wire to the **C** terminal. If another wire is attached to the **C** terminal, stop now and contact a qualified contractor for help.
4. If your old thermostat had **Y1**, **W1** and **W2** wires, stop now and contact a qualified contractor for help.
5. **C** does not power the thermostat display or operations; batteries are always required.
6. This model doesn't support the heat pump fault alert (L/A terminal). If this is desired, please contact a contractor for replacement model.

IMPORTANT: Do **NOT** use **W** for heat pump applications. Auxiliary heat must wire to **AUX** or **E**.

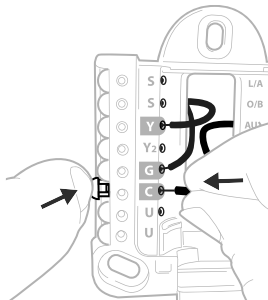


4 Connect wires to the UWP.

Refer to the notes you recorded on the chart during removal. Depress the tabs to put the wires into the inner holes of their corresponding terminals on the UWP (one wire per terminal) until it is firmly in place.

Gently tug on the wires to verify that they are secure.

Tip: If you need to release the wires again, push down the terminal tabs on the sides of the UWP.

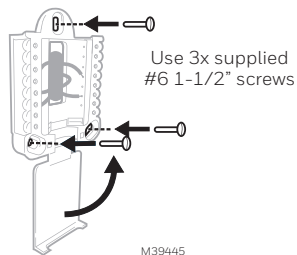


This wiring is just an example; yours may vary.

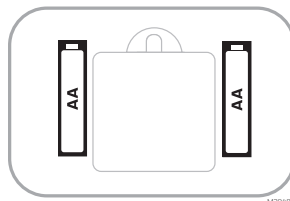
5 Confirm wiring matches snapshot.

Confirm wiring matches terminals from the photo you took during removal.

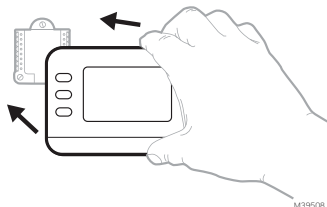
- 6 Mount the UWP and close the door.**
Mount the UWP using the provided screws. Install all three screws for a secure fit on your wall. Close the door after you're finished.



- 7 Install batteries.**
Insert two AA alkaline batteries in the back of the thermostat as shown.

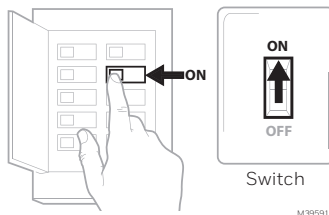


- 8 Attach the thermostat.**
Align the thermostat onto the UWP and firmly snap it into place.



- 9 Turn the power ON.**
Turn on the power at the breaker box or switch that controls the heating/cooling system.

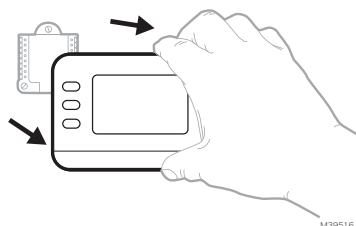
Breaker box



Battery Replacement

The display will alert you when the batteries are low and must be replaced. Remove the thermostat by pulling it away from its mount as seen at right.

Be sure to use fresh AA batteries and insert them as shown in Step 7. After inserting the new batteries, align the thermostat with the mounting plate and push gently until the thermostat snaps back into place as seen in Step 8.



Configuration

Set Time and Date

Once power is applied for installation, the date will start flashing.

Time

1. Press **MENU** on the thermostat
2. Press $\uparrow$ or $\downarrow$ to go to **TIME** and press **SELECT**
3. Press $\uparrow$ or $\downarrow$ to choose 12 or 24 hour clock format and press **NEXT**
4. Press $\uparrow$ or $\downarrow$ to adjust the hour and press **NEXT**
5. Press $\uparrow$ or $\downarrow$ to adjust the minute and press **NEXT**

Date

1. Press **MENU** on the thermostat
2. Press $\uparrow$ or $\downarrow$ to go to **DATE** and press **SELECT**
3. Press $\uparrow$ or $\downarrow$ to choose **YEAR** and press **NEXT**
4. Press $\uparrow$ or $\downarrow$ to choose **MONTH** and press **NEXT**
5. Press $\uparrow$ or $\downarrow$ to choose **DATE** and press **SAVE** and then **EXIT**

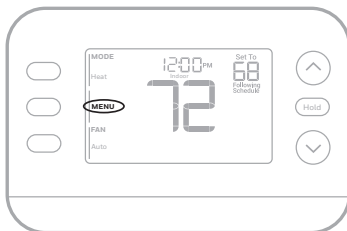
System Setup

On initial setup, the thermostat will enter the ISU menu after setting the time and date. If entering the ISU menu after initial setup, follow the steps below:

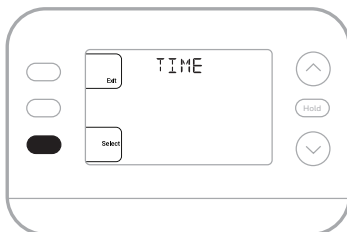
1. Press and hold the bottom left button and Up arrow $\uparrow$ button for 5 seconds to access the **INSTALLER SETUP**
2. The display shows the ISU number on the left and the ISU setting on the right.

When an ISU number is displayed, press $\uparrow$ or $\downarrow$ to change the setting for that ISU.

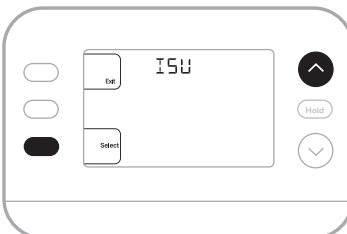
3. After choosing the correct setting for an ISU, press **NEXT** to advance to the next ISU setting.
4. To finish setup, press the **SAVE & EXIT** button; this will save your settings and return to the Home screen.



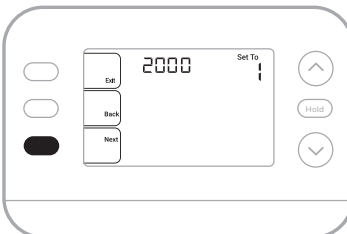
M39509



M39510



M39512



M39506

Installer Setup Options (ISU)

Depending on system settings, not all options may be available

ISU	ISU Name	ISU Options (factory default in bold)	
1040	Scheduling Options (If you do not wish to set a schedule, use the default setting, then enter the Schedule Menu after completing setup and select Off)	1 = 1 week (all days the same) 2 = 5-2 Programmable (separate weekdays and weekend) (The RTH20B only has the 5-2 day setting) 3 = 5-1-1 Programmable (separate Saturday and Sunday) 4 = 7-Day Programmable (each day separate)	
1050	Temperature Indication Scale	F = Fahrenheit	C = Celsius (<i>Default varies by model</i>)
2000	Heating System Type	1 = Conventional Forced Air Heat 2 = Heat Pump	3 = Radiant Heat (Boiler) 5 = None (Cool Only)
If you are unsure what type of system you have, refer to your heating/cooling system equipment literature or call an HVAC professional. If you select the wrong system type, the thermostat will not operate the equipment as intended.			
2010	Heating Equipment Type	Conventional Forced Air Heat: 1 = Standard Efficiency Gas Forced Air 2 = High Efficiency Gas Forced Air 3 = Oil Forced Air 4 = Electric Forced Air	5 = Hot Water Fan Coil (ISU 2010 not shown when 2000 = Heat Pump) Radiant Heat: 9 = Hot Water Radiant Heat 12 = Steam
2060	Reversing Valve O/B	0 = 0 (O/B on Cool. 0 wire is connected to the O/B terminal and controls cooling.) 1=B (O/B on Heat. B wire is connected to the O/B terminal and controls heat.) The literature that came with your heat pump equipment should indicate whether the reversing valve is energized in Heat or Cool mode	
2070	Cool Stages/Compressor Stages	0, 1 , 2 (<i>1 if 2000 = 2</i>) Only 1 compressor stage available on RTH20B/RTHC20B and RTH21B/RTHC21B. Select the number of Stages. For heat pumps, only 1 stage is supported.	
2071	Heat Stages/Backup Heat Stages	Heat Stages: 1 , 2 Backup Heat Stages: 0, 1 (RTH21B/RTHC21B and RTH22B/RTHC22B)	
2180	Aux Backup Heat Type (for Heat Pump Systems)	31=Electric Forced Air (This thermostat only supports electric backup heat. If your heat pump uses a gas or oil backup heat source, this thermostat should not be used.)	
3000	System Changeover	0 = Hidden (Manual only) 1 = Enabled (Automatic available. In auto mode, the thermostat automatically switches between heating and cooling to maintain the desired indoor temperature. This option is not recommended if the outdoor temperature often stays below 50F [10C] during winter months.)	
Continued on the next page			

Installer Setup Options (ISU) continued from the previous page

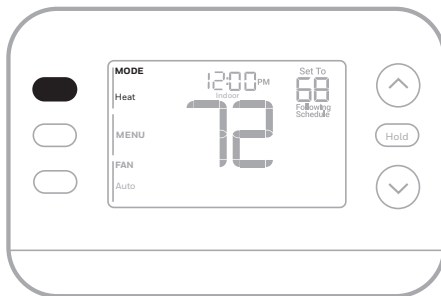
ISU	ISU Name	ISU Options (factory default in bold)	
4090	Smart Response	0=No 1=Yes (Smart Response is a comfort setting. Heating or cooling equipment will turn on earlier, ensuring the indoor temperature will match the setpoint at the scheduled time.)	
4103	Minimum Heat Setpoint	32°F to 50°F Default 40°F 0°C to 10.0°C Default 4.5 °C Do not set this lower than 40F/4.5C unless this is installed in a location where pipes are not at risk of freezing	
7110	Air Filter Replacement Reminder (Run time only counts the time that Heat, Cool or Fan are running)	0 = Off 1 = 10 Run Time Days 2 = 20 Run Time Days 3 = 30 Run Time Days 4 = 45 Run Time Days 5 = 60 Run Time Days 6 = 90 Run Time Days 7 = 120 Run Time Days 8 = 150 Run Time Days 9 = 30 Calendar Days	10 = 45 Calendar Days 11 = 60 Calendar Days 12 = 75 Calendar Days 13 = 3 Calendar Months 14 = 4 Calendar Months 15 = 5 Calendar Months 16 = 6 Calendar Months 17 = 9 Calendar Months 18 = 12 Calendar Months 19 = 15 Calendar Months
14005	Idle screen selection	0 - Minimum Information shown 1 - Setpoint shown on idle screen 2 - Maximum display information shown	
14010	Clock format	12/24	
14015	Daylight Saving Time	0=Off 1=On	

System Operation Settings

1. Press the **MODE** button to cycle to the next available System mode
 2. Cycle through the modes until the desired System mode is displayed
- Available System modes vary by model and system settings.

System modes:

- AUTO
- HEAT
- COOL
- EM HEAT
- OFF



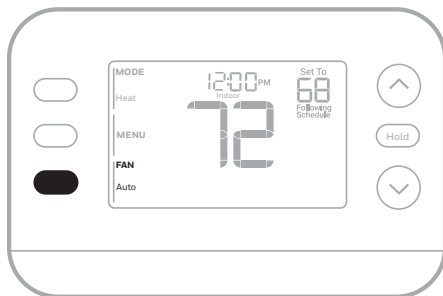
M39511

Fan Operation Settings

1. Press the **FAN** button to cycle to the next available Fan mode
2. Cycle through the modes until the desired Fan mode is displayed
Available Fan modes vary with system settings.

Fan modes:

- **AUTO:** Fan runs only when the heating or cooling system is on
- **ON:** Fan is always on
- **CIRC:** Fan runs about 33% of the time to circulate air



M39512

Program Schedule

You can program four time periods each day, with different settings for weekdays and weekends. We recommend using the presets shown in the table at right; these presets are designed to reduce your heating/cooling expenses.

WAKE: Set to the time you wake up and your desired temperature during the morning until you leave for the day.

AWAY: Set to the time you leave home and your desired temperature while you are away (usually an energy-saving temperature).

HOME: Set to the time you return home and your desired evening temperature until bedtime.

SLEEP: Set to your bedtime and your desired overnight temperature (usually an energy-saving temperature).

	Heat	Cool
Wake (6:00 am)	70°	78°
Away (8:00 am)	62°	85°
Home (6:00 pm)	70°	78°
Sleep (10:00 pm)	62°	82°

Fahrenheit Deg.

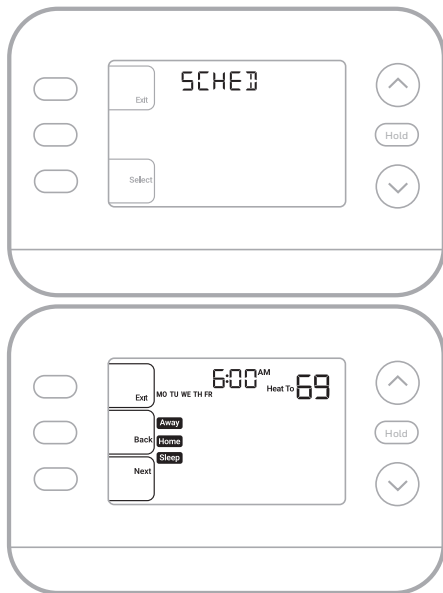
M36013

Adjusting program schedules

1. Press **MENU**. Press $\uparrow$ or $\downarrow$ until **SCHED** is displayed.
2. A square appears around **ON** or **OFF** in the display. If you want to use a schedule, press $\uparrow$ or $\downarrow$ to select **ON**. Press **Select** to edit the schedule or press **EXIT** to exit the menu.
3. If editing the schedule, press **SELECT** to edit the day or days flashing in the display.
4. The word **Wake** should be flashing. Press **NEXT**. Press $\uparrow$ or $\downarrow$ to turn this schedule period on or off. Note that selecting **Off** will disable this schedule period. Press **NEXT**.
5. If the schedule period was set to **ON**, the time will be flashing. Press $\uparrow$ or $\downarrow$ to adjust the time for this period. Press **NEXT**.
6. The Heat setpoint will be flashing. Press $\uparrow$ or $\downarrow$ to adjust the Heat setpoint for this period. Press **NEXT**.
7. The Cool setpoint will be flashing. Press $\uparrow$ or $\downarrow$ to adjust the Cool setpoint for this period. Press **NEXT**.
8. The display will show the next schedule period flashing. Repeat steps 4 – 7 for the Away, Home and Sleep schedule settings. After completing all schedule settings for the day(s) selected, repeat these steps for the other days.
9. Press the **SAVE & EXIT** button at upper left.

When editing an existing schedule, you can advance to the setting you wish to change, edit that setting and press the **SAVE & EXIT** button to save changes.

Depending on what type of schedule the thermostat was set up for, days may be grouped or set individually.



M39532

Smart Response Feature

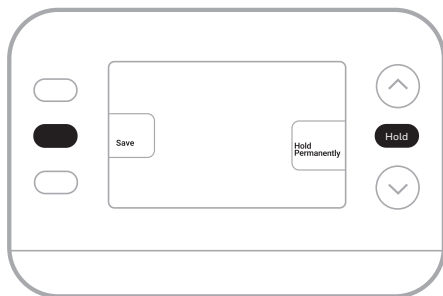
This feature allows the thermostat to “learn” how long the furnace and air conditioner take to reach programmed temperature settings, so the temperature is reached at the time you set. For example:

Set the Wake time to 6 am and the temperature to 70°. The heat will come on before 6 am, so the temperature is 70° by the time you wake at 6. The message “**Active Recovery**” is displayed when the system is activated before a scheduled time period.

Program schedule override (Temporary)

1. Press ↑ or ↓ to adjust the temperature.
2. Once at the desired setpoint temperature, no further action is needed. The new setpoint temperature will be held until the next scheduled time period begins.
3. Press the **SAVE** button or wait for the display to time out and return to the Home screen.

To cancel the Temporary Hold, press and release the **HOLD** button to cycle through the settings until **CANCEL HOLD** is selected.



Program schedule override (Permanent)

1. Press ↑ or ↓ to adjust the temperature.
2. Once at the desired setpoint temperature, press and release the **HOLD** button to cycle through the settings until **HOLD PERMANENTLY** is selected.
3. Press the **SAVE** button or wait for the display to time out and return to the Home screen.

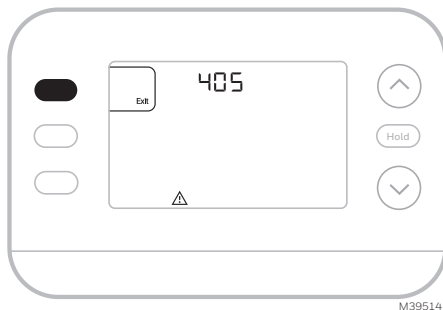
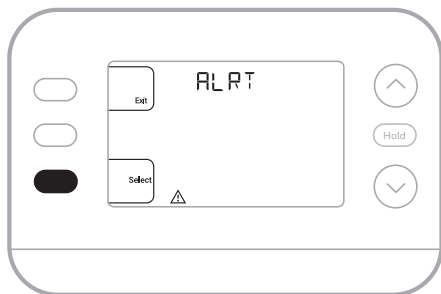
To cancel the Permanent Hold, press and release the **HOLD** button to cycle through the settings until **CANCEL HOLD** is selected.

Alerts

When an alert is active, an  icon will appear in the lower part of the display.

1. Press **MENU** and use $\uparrow$ or $\downarrow$ until the alert is shown. Active alerts will be the first menu items shown.
2. Press **SELECT** to display the alert number.
3. An alert cannot be dismissed. If there is more than one active alert, pressing the **NEXT** button allows you to view any additional alert numbers. Press **EXIT** to return to the home screen.

Some alerts can be resolved by the homeowner, for example **Replace Batteries**. Other alerts may require a service call to the professional installer.

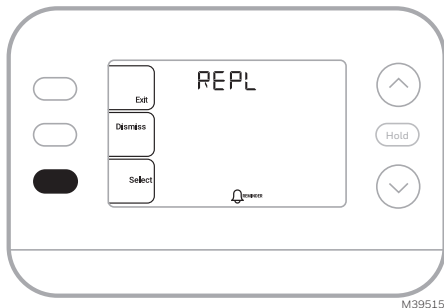


Alert Number	Alert Meaning
405	Thermostat batteries are low. Replace batteries.
407	Thermostat batteries are critically low. Replace batteries.
170	Thermostat Memory Failure. Internal problem with the thermostat memory.
171	Set Time and Date.
173	Internal Sensor Error. Issue with the built-in temperature sensor.

Reminders

When a reminder is active, a  icon will appear in the lower part of the display.

1. Press **MENU** and use $\uparrow$ or $\downarrow$ until the active reminder is shown. Active alerts and reminders will be the first menu items shown.
2. Press **SELECT** to display the reminder.
3. Once selected you will see the reminder message in scrolling text. Press **DISMISS** to reset the reminder or press **EXIT** to return to the home screen without resetting the reminder.



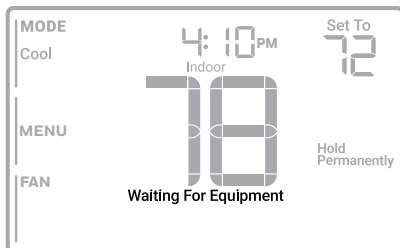
M39515

If there is more than one active reminder, press the **NEXT** button to view all reminders which have not been reset.

Built-in Compressor Protection

Damage can occur if your system's compressor is restarted too soon after shutdown. This feature forces the compressor to wait for a few minutes before restarting. During the wait time, the display will show the message **Waiting For Equipment** under the room temperature reading.

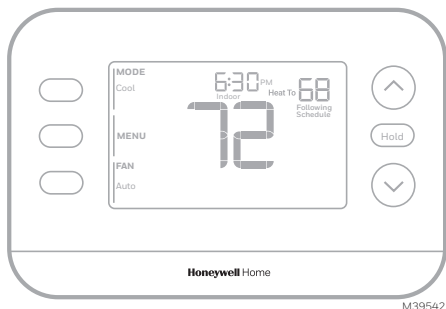
When the safe wait time has elapsed, the message disappears, and the thermostat will show "Heat on" or "Cool on".



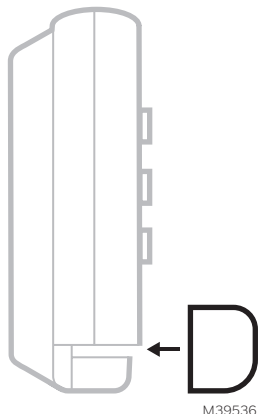
M39528

Accent Piece Replacement

One white and one grey accent piece are provided with the RTH21B/RTHC21B and RTH22B/RTHC22B thermostats to accommodate different user preferences.



To replace the accent piece, pull it away from the thermostat...



...then snap the new one on.

See the video at:



Troubleshooting

Display is blank	Make sure fresh AA alkaline batteries are correctly installed (see Step 7 in the section Intalling the Thermostat)
Heating or cooling does not run	- When running heat, display will show HEAT ON in lower right of display - When running cool, display will show COOL ON in lower right of display - If display shows WAITING FOR EQUIPMENT under temperature reading, it is in compressor delay mode to protect the system. Wait 5 minutes to see if the thermostat makes a Heat or Cool call - If display does not indicate a call for Heat or Cool or WAITING FOR EQUIPMENT, verify the mode setting, temperature setpoint and room temperature If the Up or Down arrow is pressed - Upper left of display shows mode setting - Upper right shows setpoint - Center of display shows room temperature If the issue persists - Check circuit breaker and reset if necessary - Make sure power switch at heating & cooling system is on - Make sure furnace door is closed securely
Heat or Aux Heat runs with Cooling	- Verify there is not a wire attached to W for heat pump systems. See Wiring section - For heat pump applications the reversing valve is energized in Heat on some heat pumps and Cool for other heat pumps. Verify that ISU 2060 is set correctly - Verify that no wires are shorted. Look for exposed sections of wire at the UWP
Heat or Aux heat runs with no call for heat or cooling	- Verify display does not indicate HEAT ON, COOL ON or AUX HEAT ON - Verify there is not a wire attached to W for heat pump systems. See Wiring section. - Verify that no wires are shorted. Look for exposed sections of wire at the UWP
Cannot change setpoint to desired setting	- Verify the mode setting (Heat, Cool, Auto, or Em Heat in upper left of display) The setting ranges for these modes are: - Heat or Em Heat: 32 °F to 90 °F (0 °C to 32.0 °C) - Cool: 50 °F to 99 °F (10.0 °C to 37.0 °C) If the setpoint can be adjusted, but not to the full range shown above, the thermostat may be configured for a maximum heat or minimum cool setpoint to restrict settings that are energy inefficient.
WAITING FOR EQUIPMENT shown in display under room temperature	The compressor protection feature is engaged. Wait a few minutes for the system to safely restart to avoid damaging the compressor.